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Transportation Research Procedia 78 (2024) 166–173

Transportation Research Procedia

www.elsevier.com/locate/procedia

25th Euro Working Group on Transportation Meeting (EWGT 2023)

A time-series approach for estimated time of arrival prediction in autonomous vehicles

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Abstract

Autonomous Vehicles are expected to play a pivotal role in the future of transportation. While huge accomplishments have been conducted in this field during the recent years, the main field of relevant research is primarily private transportation. Public transport (PT), especially in urban areas, although equally important, is lagging behind when the technological breakthroughs are considered. One of the main PT desiderata is punctual arrival. Therefore, this paper deals with the Estimated Time of Arrival (ETA) issue, tackled as a time-series problem, using contemporary Gradient Boosting (GB) methods, in order to benefit both commuters and stakeholders. The GB methods used are eXtreme Gradient Boosting (XGBoost), Light Gradient Boosting Machines (LightGBM), and Categorical Boosting (CatBoost). This study proposes a competitive ETA methodology for Autonomous Shuttles, providing encouraging results both in the field of Research & Innovation for Autonomous Vehicles and Public Transport, being able to be expanded for commercial purposes.

Keywords: Estimated Time of Arrival; CCAM; XGBoost; LightGBM; CatBoost; Time-series.

1. Introduction

Autonomous Vehicles (AVs) is an innovation field expected to enhance the efficiency of Public Transportation (PT). Advanced equipment, such as satellite navigation systems (e.g. GPS), digital maps, communication technologies (Smietanka et al. (2015)), along with the ability to install sensors measuring and calculating other performance indicators, will improve the offered Quality of Service from stakeholders, along with the commuter experience. Co-

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2352-1465
10.1016/j.trpro.2024.02.022

operative, Connected and Automated Mobility (CCAM)[1] is a field of special interest, especially within the European Union.

While PT plays a prominent role in contemporary European cities, user acceptance of shuttle automation is still at stake, as the results of previous studies do not provide concrete results (Bernhard et al. (2020)), (Yap et al. (2016)). Enhanced Estimated Time of Arrival (ETA) prediction, could play a pivotal role in the shift toward Autonomous Mobility. Assuming AVs are electric-powered, both the urban environment could be improved, along with the cost savings due to automation that typically exceed the AV capital cost. European cities pay extra attention to PT, highlighting the importance of the precision of the means of transportation (Luethi et al. (2007)). Accuracy of precision creates trust in public transport, while ETA manifests its prominence, by both facilitating PT providers to ensure reliability and passengers to reach their destination on time.

The proposed study investigates ways to approach ETA prediction as a time-series problem, where the arrival from point A to point B (i.e. from one shuttle stop to the next) takes the previous time steps into calculation apart from the current state. The goal is to provide a reliable service to commuters, that takes into consideration previous characteristics contributing in latency, to calculate the exact time passengers board the shuttle and reach their destination.

The remainder of this study is structured as follows. In Section 2, some previous related studies are presented, while in Section 3, the methodology for data acquisition and processing is discussed, along with the nominated GB algorithms used in this approach. Finally, in Section 4, the results of this research are analyzed, while in Section 5, a conclusion is presented, discussing the overall achievements of this approach and proposing further work for the future.

2. Related work

For the purpose of this study, a number of previous works were taken into consideration, mainly from the time period of 2015 to 2022. The main interest of all these papers is the calculation of ETA using Gradient Boosting (GB) methods among others, for various means of transportation (train, airplanes, cars, taxis, buses). The common outcome of all these studies is that GB approaches (XGBoost, LightGBM, CatBoost) typically outperform other regression methods (statistical approaches, Neural Networks, LSTMs, Random Forest etc.).

Predicting the arrival time of shuttles in PT is a research field with surged interest during the last decades. A number of approaches have been implemented so far, however the emergence of automated buses have increased the complexity of this field. Several Machine Learning (ML) approaches, such as Neural Networks (NNs), Long Short Term Memory (LSTM) networks, Support Vector Machines (SVMs) and K-Nearest Neighbors (KNNs), have been substituted with GB approaches, such as XGBoost, CatBoost and LightGBM, using the previous as benchmark. Massive transportation, either by airplanes, trains, buses or even cars and taxes, requires punctual and reliable ETA calculation, and therefore, regardless of the transportation means, ETA calculation remains of utmost importance.

A number of studies in the literature have utilized GB techniques as part of their methodologies. Among others, the field of GB algorithms has been researched in Taparia and Brady (2021) for bus arrival time prediction, in Dou (2020), Wang et al. (2020), and Yang et al. (2020) for flight transportation, while train transportation has been researched in Li et al. (2021). Moreover, urban and peri-urban PT is of equal importance, either considering bus scheduled itineraries or demand responsive transport.

ETA prediction is a valuable tool and can also be used in logistics & supply chain as having been performed by Poschmann et al. (2023). This study presented how various ML methods, including XGBoost, could enhance the ability to predict the arrival time of cargo, using multiple transport means. The mindset of this study could also be incorporated in urban PT, as a number of commuters use multiple transportation methods and could benefit from a multi-modal transport approach. Similarly, a multi-modal approach met in the literature dealing with the route optimization problem utilized Graph NNs (Liu et al. (2023)).

When public transport is considered, a representative research work in the ETA field is that of Reich et al. (2020). In this research, a bus route in Bournemouth (UK) is researched for ETA calculation, while the main ML methods used are Recurrent NNs, LSTM and Gated Recurrent Units, with stop distances of 200 meters. Another ML model,

[1] Cooperative, Connected Autonomous Mobility, https://www.ccam.eu/, Accessed: 2023-02-23

named DeepTRANS (Tran et al. (2020)), have been proposed for the purpose of calculating Bus ETA in Los Angeles, a Diffusion Convolutional Recurrent NN, utilizing both a Geo-Convolution layer and an LSTM layer.

The status of integration on AVs in PT is as follows. User Acceptance has been one of the first field researched (Pakusch and Bossauer (2017)), where most users were positive toward owning and/or using autonomous cars and shared AVs, such as the ones used in PT that can replace up to 11 conventional vehicles. In fact, according to this paper, user acceptance in automation of PT is higher than automation in private transportation. Moreover, PT automation, especially Autonomous Mobility-on-Demand (Salazar et al. (2018)), (Distler et al. (2018)), can be optimized in various ways in order to solve critical issues in urban transportation, such as traffic congestion, energy consumption and CO2 emissions, while simultaneously increasing the commuters Quality of Experience and sense of Security and Trust in automated mobility.

Cost optimization for commuters is a serious issue in PT as well. Self-driving technologies enable substantial cost reduction for all transportation modes (Bösch et al. (2018)), with the autonomous city bus being the best cost-effective option for mass transportation. Hence, automation in PT can become competitive to vehicle ownership in terms of cost, once commuting time and comfort is taken into account. Shared Autonomous Mobility is concurrently, the most equitable transportation method (Whitmore et al. (2022)), as provides transit coverage for substantially, lower costs than conventional transportation methods, along with increased access to currently under-served areas, to satisfy modern needs for all-inclusive commuting.

In order for AVs to function seemlessly within PT, a number of services need to be implemented to provide high quality of service to commuters. An initial preliminary research work has been provided in Antypas et al. (2022), to deal with the burgeoning adaption of CCAM. The main focus of this work is to calculate ETA using contemporary ML techniques. XGBoost, CatBoost and LightGBM are all taken into consideration, using RSME, MAE and MAPE as evaluation metrics, in order to predict ETA. Moreover, this paper proposed a methodology for data pre-proccessing, using the haversine distance formula, in case there are no set stations within an itinerary. The proposed algorithm could also be used for Demand Responsive Transport, along with PT solutions. The research work of Antypas et al. (2022) managed to produce encouraging results, achieving an around of 30% percentage error.

From all the aforementioned bibliography, the lack of autonomous shuttle incorporation in the ETA literature is apparent. Both because the field on AVs is not fully integrated within urban PT, and because the research considering AVs is mainly focused on privately owned cars, not buses. Therefore, to the best of our knowledge, the field of AVs in PT is still germinal, where ML technologies have not yet been fully incorporated, while the research community is mainly occupied with user acceptance and the impact of such shuttles in PT systems. The current literature, therefore, still needs to make a great leap forward towards the incorporation of CCAM services in AVs, such as ETA prediction, along with a number of other Mobility-as-a-Service solutions. The proposed methodology expects to go beyond the previous related work of the field by approaching ETA using time-series consideration, and this is the main novelty and differentiation of the current work in comparison with the work of Antypas et al. (2022). Essentially, the present research work can be considered a continuation of the previous research study, building upon it.

3. Methodology

3.1. Data Preproccessing

For the purpose of this study, data were acquired from a pilot site within the SHOW project ecosystem. The collected data consist of the AV's timestamp, location, speed and acceleration, as well as the outdoor weather. The collected timestamps where converted to date, time of the day and cumulative seconds within each day. Data were acquired with an MQTT connection, where the AV transmission is one entry per second, for location, timestamp, speed and acceleration, while the weather transmission is sparse and irregular (only when the AV detects changes).

The first challenge of this study is to calculate the distance between two specific points, without considering the fixed shuttle stops. Although the respective literature (Tirachini (2014)) suggests urban PT stops should not exceed 300 meters, 400 meters between the stops are used in real-life itineraries. Thus, the considered scenarios for the proposed methodology are stops with 100, 200, 300 and 400 meters distance between them, following the consideration of Antypas et al. (2022).

Inspired by the aforementioned study, the proposed methodology uses the same algorithmic schema for the data prepossessing needed to create the final dataset. The data pre-processing includes the calculation of the distance between the desired stops, the different latency scenarios consideration, and the zoning method for each day of the week and time of the day. More particularly, the proposed algorithm utilizes the haversine formula in order to calculate the distances. Furthermore, in order to take latency into account, two scenarios were considered, where 10 minutes latency (first scenario) or 20 minutes latency (second scenario) were regarded as noise outliers and therefore were removed from the dataset. Finally, a zoning method to classify days of the week and time zones within the day were considered. A number was appointed to each week day (from 1 to 7), while entries from the earliest to the statistical average were classified as morning, and entries from the statistical average to the latest were classified as afternoon, with the last classification corresponds to no shuttle operation.

The final dataset was treated as a time-series dataset and as mentioned above, this is the main differentiation of the proposed algorithm in comparison with the related work of Antypas et al. (2022). Therefore, previous time steps were taken into consideration, i.e the time needed to cover the desired distance for previous stops within the itinerary. The following scenarios were considered. One where no previous time was taken into account t, and other four where the GB algorithms use incrementally in addition to the current state, the time needed for the four last stops $(t-1)$ to $(t-4)$, compiling a methodology with five instances for every distance-latency pair, in the context of this research. The final equation for the formed problem is presented in equation (1)

$$y(t) = func\left(x_1, x_2, x_3, x_4, x_5, x_6, y(t-s)_{\text{for } s=1 \text{ to } 4}\right) \tag{1}$$

In equation (1), $func()$ is the used Gradient Boosting function, x_1, x_2, x_3, x_4, x_5, x_6 represent the speed, acceleration, location, weather, day of week, and time of day variables, while $y(t-s)_{\text{for } s=1 \text{ to } 4}$ is the loop of the previous time steps taken into consideration (from the preceding to the last, up to 4 time steps).

3.2. Gradient Boosting Algorithms

For the context of this paper, three GB methods were selected, in accordance with the related work (Antypas et al. (2022)), to focus on the generalization of this work. The purpose of this study is not to provide a comparison between these three algorithms, but to ensure that these results are not constricted only within the boundaries of one approach. The three GB algorithms used in this analysis, are the following:

- **eXtreme Gradient Boosting (XGBoost)** (Chen and Guestrin (2016)). A scalable end-to-end tree boosting system to minimize loss function, in order to build an additive expansion of the objective function.
- **Light Gradient Boosting Machines (LightGBM)** (Ke et al. (2017)). A free and open-source extensive library to implement GB methods with extensive variants. Gradient-Based One Side Sampling (GOSS) takes advantage of the absence of native weight for the data instances in GB Decision Trees.
- **Categorical Boosting (CatBoost)** (Prokhorenkova et al. (2018)). A method that addresses the problem of prediction shift that occurs when GB algorithms use the same instances for estimating both the gradients and the models that minimize those gradients (target leakage).

Finally, according to the related work presented in Section 2, the evaluation metrics that were consistently used are: Root Mean Square Error (RMSE), Mean Absolute Error (MAE) and Mean Absolute Percentage Error (MAPE). Therefore the aforementioned metrics are utilized in this study as well, along with the coefficient of determination R-squared (R^2).

4. Results

For the research purpose, a total of eight research scenarios is presented, each with 5 time steps considered (from only the current time to four previous time steps), constructing 40 different research approaches. The first two scenarios considered, as presented in table 1, consist of desired distances of 100 meters between two points, where latencies exceeding 10 minutes or 20 minutes were omitted from the final dataset. In accordance with the analysis in subsection 3.1, all the time step instances are presented in a comparative manner. The main goal is to present the benefits of handling such problems as time series, while also discovering the optimal solution considering the number of previous time steps needed for sufficient results. Using the same premise, tables 2, 3 and 4 were constructed.

Table 1. Comparative results for the 100 meter scenarios for all time steps.

		10 minutes latency					20 minutes latency				
Algorithm	Evaluation Metrics	t	$(t-1)$	$(t-2)$	$(t-3)$	$(t-4)$	t	$(t-1)$	$(t-2)$	$(t-3)$	$(t-4)$
XGBoost	RMSE (s)	48.12	13.10	16.57	16.90	16.01	53.70	27.34	17.22	20.16	14.69
	MAE (s)	22.21	1.57	2.37	5.21	1.85	26.61	2.26	1.91	3.51	1.73
	MAPE (%)	32.01	1.93	1.60	9.52	2.80	45.46	2.49	2.53	0.98	2.70
	R^2	0.32	0.94	0.96	0.90	0.91	0.15	0.87	0.94	0.96	0.96
LightGBM	RMSE (s)	51.12	15.90	22.90	14.77	14.97	74.67	19.01	16.83	28.30	14.83
	MAE (s)	28.08	4.54	4.82	2.61	2.84	45.01	5.63	3.84	16.42	2.97
	MAPE (%)	48.52	9.77	4.48	4.67	5.55	95.47	12.62	7.90	6.55	6.57
	R^2	0.23	0.91	0.92	0.92	0.92	0.64	0.94	0.94	0.93	0.96
CatBoost	RMSE (s)	49.32	11.30	18.64	12.49	13.16	48.88	11.94	12.09	22.94	11.65
	MAE (s)	22.06	1.40	2.91	1.54	1.55	27.28	1.57	1.46	4.54	1.44
	MAPE (%)	31.90	1.67	2.00	1.95	2.10	50.13	2.12	2.00	1.52	2.07
	R^2	0.29	0.96	0.95	0.94	0.94	0.30	0.97	0.97	0.95	0.97

Table 2. Comparative results for the 200 meter scenarios for all time steps.

		10 minutes latency					20 minutes latency				
Algorithm	Evaluation Metrics	t	$(t-1)$	$(t-2)$	$(t-3)$	$(t-4)$	t	$(t-1)$	$(t-2)$	$(t-3)$	$(t-4)$
XGBoost	RMSE (s)	51.73	30.32	18.79	17.43	17.32	79.06	16.13	15.76	25.54	24.20
	MAE (s)	31.35	16.30	2.80	2.73	2.49	39.01	2.59	1.87	3.39	3.29
	MAPE (%)	29.33	14.00	1.88	1.94	1.62	41.52	1.76	2.65	3.21	2.85
	R^2	0.59	0.86	0.94	0.95	0.95	0.04	0.98	0.95	0.95	0.95
LightGBM	RMSE (s)	61.88	24.87	21.82	24.85	22.73	90.41	35.08	16.17	28.78	29.25
	MAE (s)	39.48	13.83	4.67	4.79	4.52	63.81	8.16	3.64	6.34	5.97
	MAPE (%)	33.22	12.81	4.07	4.00	3.78	61.21	7.52	7.44	6.22	5.30
	R^2	0.41	0.91	0.92	0.90	0.92	0.25	0.91	0.95	0.93	0.93
CatBoost	RMSE (s)	57.02	34.23	18.06	22.86	22.43	69.30	23.67	12.71	35.99	27.84
	MAE (s)	33.55	17.56	3.99	3.36	3.08	44.24	3.73	1.49	4.07	3.88
	MAPE (%)	29.78	13.73	2.97	2.29	1.88	44.31	2.55	1.91	2.67	3.27
	R^2	0.50	0.82	0.95	0.92	0.92	0.26	0.96	0.97	0.89	0.94

In tables 1, 2, 3, and 4, the third and eighth columns (representing t) are essentially the same results having been presented in Antypas et al. (2022). The utilization of time-series approach seem to improve the results significantly. For example, regarding the strict evaluation metric RMSE in all scenarios (latency-distance-algorithm) its value has dropped dramatically from almost two minutes to a little over 20 seconds. Similarly, the value of MAE has improved clearly from over one and a half minute to 15 seconds. Continuing with the relative metrics, MAPE fell from over 80% to under 10%. Finally with respect to coefficient of determination, the time-series approach is once again superior to the only current state consideration, since R^2 presents for almost all scenarios values over 0.9, which are particularly higher in comparison with the simple current state approach. It is important to note that the more time steps used, the more prone to overfitting the algorithm is due to the variable space increase.

Regarding the comparison between the various scenarios of previous time steps $(t-1)-(t-4)$, the most radical performance increase in the present research, seems to be when using one $(t-1)$ or two $(t-2)$ previous time steps. The R^2 metric drastically increases when even one previous time step is considered, but does not seem to increase when

Table 3. Comparative results for the 300 meter scenarios for all time steps.

		10 minutes latency					20 minutes latency				
Algorithm	Evaluation Metrics	t	$(t-1)$	$(t-2)$	$(t-3)$	$(t-4)$	t	$(t-1)$	$(t-2)$	$(t-3)$	$(t-4)$
XGBoost	RMSE (s)	78.25	30.18	23.05	24.17	25.35	94.26	33.31	19.71	18.57	18.01
	MAE (s)	48.34	4.33	3.95	4.29	4.66	57.79	5.39	3.61	3.13	3.18
	MAPE (%)	31.48	1.60	1.60	1.82	1.97	38.10	2.19	1.81	1.30	1.47
	R^2	0.47	0.95	0.93	0.92	0.92	0.23	0.94	0.98	0.98	0.98
LightGBM	RMSE (s)	80.26	35.72	23.14	26.37	24.16	96.45	24.98	24.15	27.95	29.28
	MAE (s)	51.92	19.12	7.09	7.03	7.28	69.32	10.49	7.66	8.33	8.26
	MAPE (%)	24.68	11.45	3.83	3.75	3.65	46.17	6.05	4.34	4.58	5.02
	R^2	0.44	0.94	0.93	0.91	0.92	0.19	0.97	0.97	0.96	0.95
CatBoost	RMSE (s)	94.84	24.01	22.06	23.76	23.59	90.72	21.71	20.58	21.48	20.65
	MAE (s)	60.81	4.28	3.90	3.79	4.28	57.82	3.60	3.64	3.99	3.87
	MAPE (%)	35.77	1.97	1.79	1.49	1.86	38.18	1.62	1.72	1.88	1.96
	R^2	0.22	0.97	0.94	0.93	0.93	0.29	0.98	0.98	0.97	0.98

Table 4. Comparative results for the 400 meter scenarios for all time steps.

		10 minutes latency					20 minutes latency				
Algorithm	Evaluation Metrics	t	$(t-1)$	$(t-2)$	$(t-3)$	$(t-4)$	t	$(t-1)$	$(t-2)$	$(t-3)$	$(t-4)$
XGBoost	RMSE (s)	78.43	21.08	21.46	20.29	20.35	104.93	21.40	20.80	20.49	22.49
	MAE (s)	50.83	3.85	3.98	3.74	3.99	69.01	3.88	3.46	3.77	3.66
	MAPE (%)	28.67	1.12	1.20	1.05	1.10	35.66	1.10	0.98	1.04	0.97
	R^2	0.59	0.96	0.96	0.96	0.96	0.38	0.98	0.98	0.98	0.97
LightGBM	RMSE (s)	85.11	27.73	25.02	24.56	25.95	141.59	37.97	43.32	42.37	33.68
	MAE (s)	57.85	18.01	14.11	12.54	12.84	106.18	10.30	12.54	11.23	7.92
	MAPE (%)	35.23	7.48	5.84	5.23	5.32	72.47	4.45	5.28	4.67	2.75
	R^2	0.52	0.94	0.94	0.95	0.94	0.14	0.94	0.90	0.91	0.94
CatBoost	RMSE (s)	74.35	22.65	24.88	23.66	22.03	133.36	20.80	39.31	36.94	20.39
	MAE (s)	51.01	4.47	4.52	4.43	4.28	92.68	4.16	7.68	7.32	3.95
	MAPE (%)	28.85	1.37	1.27	1.39	1.50	64.54	1.21	2.23	2.06	1.35
	R^2	0.63	0.96	0.94	0.95	0.96	0.10	0.98	0.92	0.93	0.98

more are considered. Moreover, it is worth-mentioning that the final time step, i.e the $(t-4)$ study in all scenarios, does not improve the yielded results. The order of magnitude is usually the same as in the $(t-2)$ and $(t-3)$ studies. The (t − 3) scenarios sometimes follow this trend, depending on the algorithm and the metric taken into consideration. For these specific scenarios and for the given dataset, taking two previous time steps into consideration would be sufficient, as the other time steps do not significantly enhance the performance of the algorithms and it is preferable to keep the variable space as minimum as possible. Therefore, one can conclude that dealing with time-series datasets in Automated Vehicles for ETA calculation, up to two previous time steps can significantly enhance the yielded results. In Figure 1, the comparison for the (t-2) scenario is presented. Before the comparison of the different scenarios it is prominent to mention that the MAPE for all the scenarios is extremely low, highlighting that way the high performance of the proposed methodology. Moreover, it is obvious from the figure, that there is a trend of minimizing the error the stricter the latency penalty gets. Similarly, when the stop distance is increased (from 100 meters to 400), the MAPE usually follows a downward trend for almost all the algorithms and scenarios.

Finally, it is important to note that, even though this is not the purpose of this research, that LightGBM consistently yields the worst results, with XGBoost and CatBoost outperforming one another, depending on the specific scenario (CatBoost is usually better when excluding latencies over 20 minutes, while XGBoost performs better with 10-minute

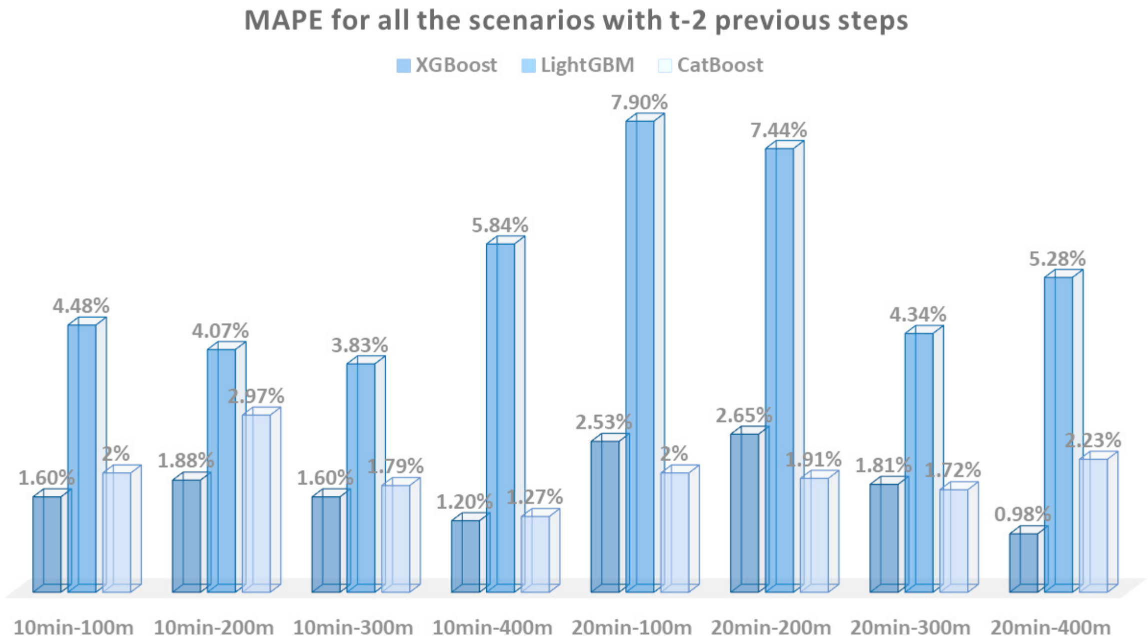

Fig. 1. XGBoost-LightGBM-CatBoost Comparison

latencies exclusion). However, this does not mean that future studies should exclude LightGBM from their methodology, in similar ETA problems. Instead, in order to be safer the generalization of similar outcomes, much more research should be performed in the future containing many different data sets.

5. Conclusion and Future Work

This paper aims to research the impact of AVs in PT. The field of Public Transportation is still under-researched, although automation in mobility and AVs are two emerging research fields. Therefore, the continuation on research and service creation on this field, enhance the Quality of Experience for commuters that prefer to travel without using a private vehicle. The goal of every itinerary is punctual arrival to the desired destination, highlighting the importance of calculation of Estimated Time of Arrival. Usage of contemporary GB algorithms, namely XGBoost, LightGBM and CatBoost, is able to produce encouraging results for further study. The case of time series approach improved the acquired outcomes dramatically compared to a previous related work that does not have time-series approach.

All the data of this study were acquired by an autonomous shuttle, operating in real-life scenarios. The encouraging results of this research justify the potential of ETA calculation in PT. However, as a future work a certain threshold needs to be defined for the optimal previous time steps. Thus, a smart system that stops training when the results do not improve significantly, therefore defining the optimal time steps and avoiding overfitting, along with the generalization of this method with bigger datasets and for various pilot sites, are of primary importance for the future of this study.

Future work could also include a real-time notification system of arrival time and the feedback in case of an unexpected event, helping with the record exclusion from the ETA calculation. Finally, more complex scenarios could be considered in the future, with higher traffic and latency, different time zones and public transportation demand affecting the traffic. Thus, bus acceleration, deceleration and speed along with shuttle dwell time in bus stops are also important factors. Furthermore, mixed traffic situation and interaction with obstacles and other objects pose an impediment to be taken into consideration. For all these reasons, more algorithms could be also considered in future research, for example including simpler and faster algorithms as benchmark, along with the combination of sophisticated Machine Learning and Deep Learning algorithms.

Acknowledgment

This work was funded by the European's Union Horizon 2020 Research and Innovation Program through SHared automation Operating models for Worldwide adoption (SHOW) project under Grant Agreement No. 875530.

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